An Introduction to Ergodicity Economics: Markets (II)

Emilie Rosenlund Soysal

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Markets

Economic agents can maximise their wealth over time by maximising the time-average growth rate of the wealth.

- What does that mean for the portfolio choice?
  - Optimal leverage
- What does that mean for the way markets organise?
  - Leverage efficiency

Optimal leverage

For a portfolio consisting of one risky asset (s) and one risk-free asset (r), the optimal share of wealth invested in the risky asset is:

$$\ell_{opt} = \frac{\mu_s - \mu_r}{\sigma_s^2} \tag{1}$$

Leverage efficiency:
Markets self-organise around

$$\ell_{opt} = 1$$

Leverage efficiency

Why $\ell_{opt} = 1$?

- Efficiency: It is impossible for an individual investor to beat the market simply by using leverage.
- Stability:
  - $\ell_{opt} > 1$: Everyone wants to borrow cash to buy stocks, but noone wants to lend money.
  - $\ell_{opt} < 0$: Everyone wants to borrow stocks and sell them to get cash, but noone wants to lend stocks.

Market simulation

The market consist of two agents, A and B,
one risky asset (s: GBM with exogenous parameters μ_s, σ_s)
and one risk-free asset (r: endogenous drift μ_r)

- A observes the risk-free rate, μ_r, and the stock dynamic, μ_s, σ_s
- A chooses its leverage, $\ell_A = \frac{\mu_s - \mu_r}{\sigma_s^2}$
- B accepts borrowing / lending at the risk-free rate according to A's needs.
- Portfolio outcomes are realised
- B adjusts the risk-free rate, μ_r

Market simulation

B's rate setting rule:

- If B's return was at least as good as A's during dt:
  - Do nothing
- If B's return was worse than A's during dt:
  - If A borrowed from B: Increase μ_r with $\Delta\mu dt\mu$
  - If B borrowed from A: Decrease μ_r with $\Delta\mu dt$

A and B start out with the same wealth.

Predictions

Does $\ell_{opt} = 1$ hold for real markets?

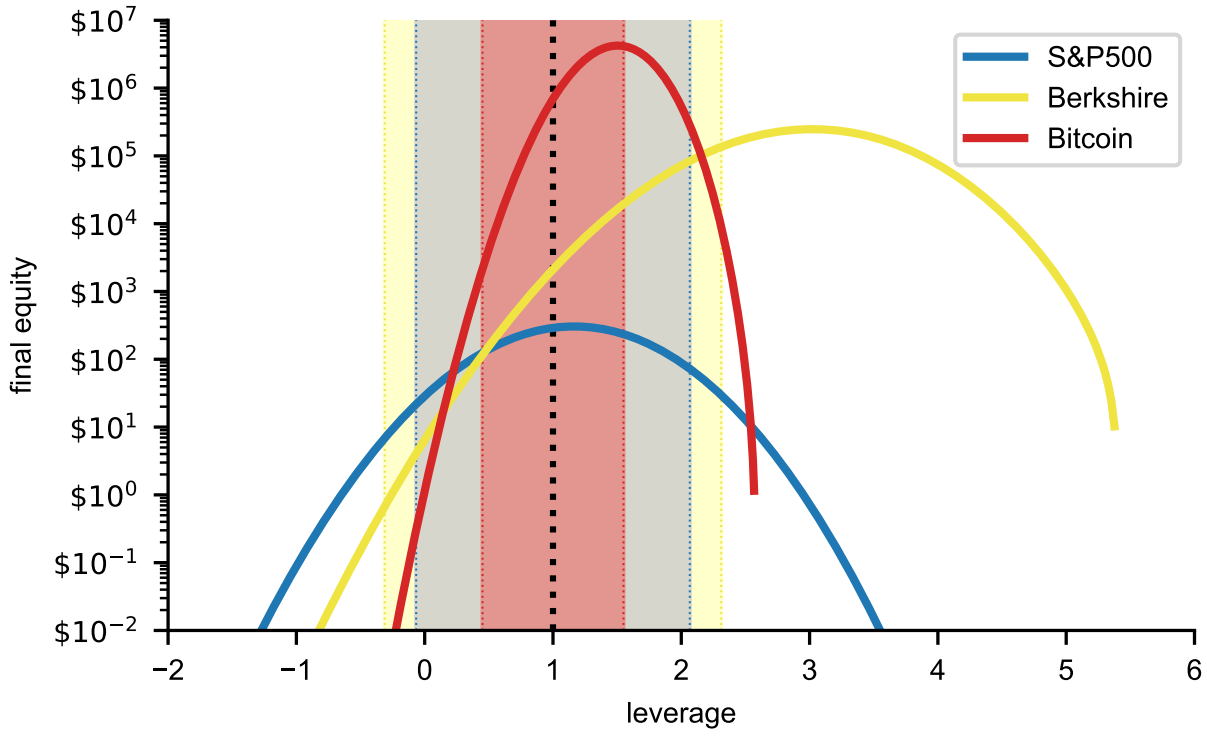

Figure 10.3 in Peters and Adamou (2025). Confidence bands indicate two standard errors of the prediction, where the standard error is $\sigma_{\ell_{opt}(\Delta t)} = \frac{1}{\sigma_s \Delta t^{1/2}}$.

Equity Premium Puzzle

Standard economic theory cannot explain the high premium on equity, but leverage efficiency can:

- $3 - 10\%$ in real life
- $0.1 - 0.5\%$ with standard theory (CCAPM)
- $0.4 - 6.4\%$ with optimal leverage ($s =$S&P500 and $r =$ FED overnight interbank rate)

Premium: difference in time average growth rate (ϵ)

$$\epsilon = \bar{g}_s - \bar{g}_r \tag{2}$$

$$= (\mu_s - \mu_r) - \frac{\sigma_s^2}{2} \tag{3}$$

$$= \frac{\mu_s - \mu_r}{2} \tag{4}$$

using leverage efficiency $\ell_{opt} = 1 \implies \mu_s - \mu_r = \sigma_s^2$

Summary

- For a risky and risk-free asset, the growth-rate optimal leverage is $\ell_{opt} = \frac{\mu_s - \mu_r}{\sigma_s^2}$
- Leverage efficient markets self-organise around $\ell_{opt} = 1$
- Leverage efficiency predicts a fixed relationship between drift, volatility and risk premium, supported by real life data

Thank you.